Appendix A

James Lee

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1. Show that the definition of rational equivalence in §1 is equivalent to the equivalence relation generated by the following relation: two cycles Y, Z of codimension r on X are equivalent if there exists a cycle W of codimension r on $X \times \mathbb{A}^1$ which intersects $X \times \{0\}$ and $X \times \{1\}$ properly, and such that $Y = W.(X \times \{0\})$, $Z = W.(X \times \{1\})$.

2. Prove the following result about Weil divisors, which generalizes (IV, Ex. 2.6), and which is needed to show that f_* is well-defined modulo ratonal equivalence (A3): Let $f : X \to X'$ be a proper, generically finite map of normal varieties, and let D_1 and D_2 be linearly equivalent Weil divisors on X. Then f_*D_1 and f_*D_2 are linearly equivalent Weil divisors on X'.

3. Show directly that any subvariety of degree d in $\mathbb{P}^n$ is rationally equivalent to d times a linear space of the same dimension, by using a projection argument similar to (III, 9.8.3).

4. Let $\pi : X \to C$ be a ruled surface (V, §2) over a nonsingular curve C. Show that the group $A^2(X)$ of zero-cycles modulo rational equivalence is isomorphic to $\operatorname{Pic} C$.

5. Let X be a surface, let $P \in X$ be a point, and let $\pi : \tilde{X} \to X$ be the monoidal transformation with center P (V, §3). Show that $A(\tilde{X}) \cong \pi^* A(X) \oplus \mathbb{Z}$, where $\mathbb{Z}$ is generated by the exceptional curve $E \in A^1(X)$, and the intersection theory is determined by $E^2 = -\pi^*(P)$.

6. Let X be a nonsingular projective variety of dimension n, and let $\Delta \subset X \times X$ be the diagonal. Show that $c_n(\mathcal{T}_X) = \Delta^2$ in $A^n(X)$, under the natural isomorphism of X with Δ.

7. Let X be a nonsingular projective 3-fold, with Chern classes c_1, c_2, c_3. Show that

$$1 - p_a = \frac{1}{24} c_1 c_2,$$

and for any divisor D,

$$\chi(\mathcal{O}_X(D)) = \frac{1}{12} D.(D - K).(2D - K) + \frac{1}{12} D.c_2 + 1 - p_a.$$

8. Let $\mathcal{E}$ be a locally free sheaf of rank 2 on $\mathbb{P}^3$, with Chern classes c_1, c_2. Since $A(\mathbb{P}^3) \cong \mathbb{Z}[h]/(h^4)$, we can think of c_1 and c_2 as integers. Show that $c_1 c_2 \equiv 0 \mod 2$.

9. *Surfaces in* $\mathbb{P}^4$

(a) Verify the formula of (4.1.3) for the rational cubic scroll in $\mathbb{P}^4$ (V, 2.19.1).

(b) If X is a K3 surface in $\mathbb{P}^4$, show that its degree must be 4 or 6.

(c) If X is an Abelian surface in $\mathbb{P}^4$, show that its degree must be 10.

(d) Determine which of the rational ruled surfaces X_e, $e \geq 0$ (V, §2), admit an embedding in $\mathbb{P}^4$.

10. Use the fact that the tangent sheaf of an Abelian variety is free, to show that it is impossible to embed an Abelian 3-fold in $\mathbb{P}^5$.